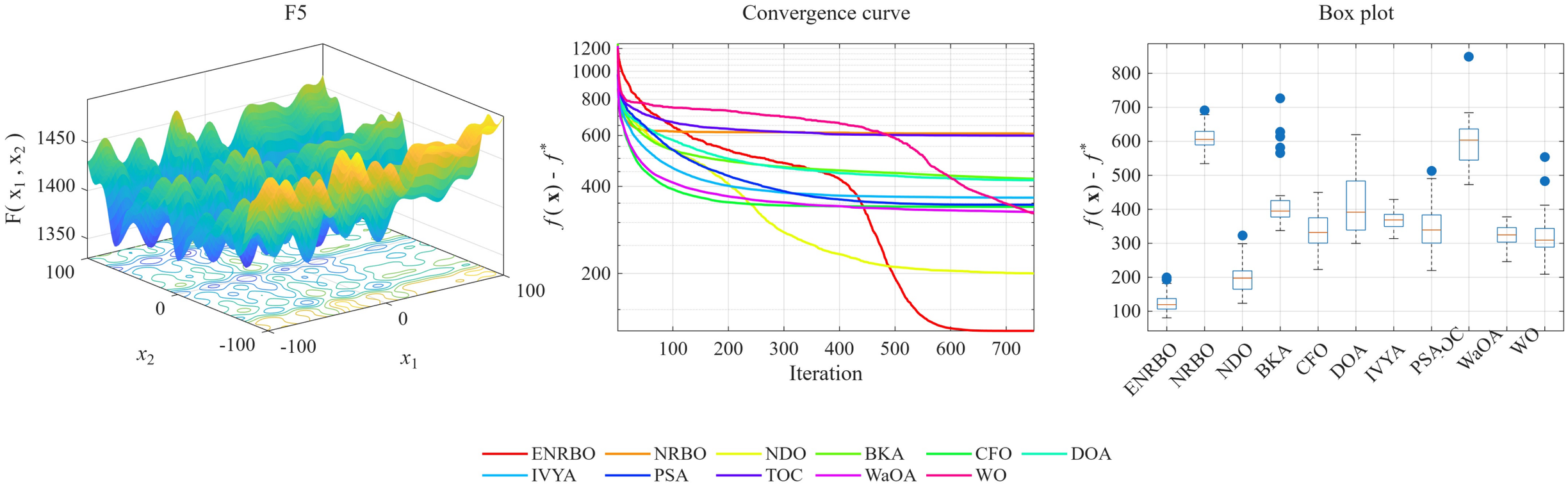

F5
F(x1 , x2)
1450
1400
1350
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
1200
1000
800
600
400
200
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
800
700
600
500
400
300
200
100
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO